Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 29)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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6	
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8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 1 . 1 Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

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0 1 . 2 Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

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0 1 . 3 Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

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0 2 . 1 Explain why compressed air energy storage with heat recovery can achieve over 90% efficiency.

[2 marks]

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0 2 . 2 Explain why large-volume hot water stores lose heat more slowly per unit of stored energy than small ones.

[2 marks]

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0 2 . 3 Explain why molten salt rather than water is used to store heat from CSP plants.

[2 marks]

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0 2 . 4 Explain how high thermal mass building materials reduce energy use.

[2 marks]

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0 2 . 5 Describe **one** other energy storage technology.

[2 marks]

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0 3 . 1 Explain the difference between nuclear fission and nuclear fusion.

[2 marks]

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0 3 . 2 State the sources of the two fuels used in fusion research.

[2 marks]

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0 3 . 3 Explain why very high temperatures, a vacuum and a magnetic field are needed in a toroidal (torus) fusion reactor.

[3 marks]

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0 3 . 4 Give **two** potential advantages of fusion over fission and one reason why it is not yet a commercial energy source.

[3 marks]

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0 4 . 1 Complete Table 1 by naming the most suitable statistical test for each hypothesis.

[4 marks]

Table 1

Hypothesis	Statistical test	
Noise levels decrease as distance from a road increases		
Fewer cars use a road after road charging is introduced (counts on 12 days before and after)		
Woodlands managed in different ways have different numbers of dormice		
Mean lobster mass is higher in areas where collection is banned		

0 4 . 2 Explain why the chi-squared test cannot be used to compare the dissolved oxygen concentrations of two rivers.

[2 marks]

0 4 . 3 Explain why a scientific investigation never 'proves' a hypothesis.

[2 marks]

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0 4 . 4 Describe how error bars are added to a graph of means and how they are interpreted.

[2 marks]

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0 5 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

A Aerobic decomposition produces little methane and returns nutrients to the soil ☒

B It produces more electricity ☐

C It removes heavy metals ☐

D It destroys asbestos fibres ☐

0 5 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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0 5 . 3 Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

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0 6 A student determined the water and organic matter content of a soil sample. Table 1 shows the results.

Table 1

Measurement	Mass / g
Empty crucible	25.0
Crucible + fresh soil	65.0
Crucible + soil after drying at 105 °C to constant mass	55.0
Crucible + soil after heating in a furnace at 550 °C	52.6

0 6 . 1 Calculate the percentage water content of the fresh soil.

[2 marks]

Answer ..... %

0 6 . 2 Calculate the percentage organic matter content of the dry soil.

[2 marks]

Answer ..... %

0 6 . 3 Explain why the soil is dried to constant mass before the organic matter is burnt off.

[2 marks]

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0 6 . 4 Explain how the organic matter content affects the fertility and erosion risk of this soil.

[3 marks]

0 6 . 5 Suggest **one** reason why two students measuring the same field might get different organic matter results.

[1 mark]

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0 7 . 1 Describe the process of anaerobic sludge digestion and give **two** benefits of the process.

[4 marks]

0 7 . 2 Explain **two** problems of using sewage sludge as an agricultural fertiliser and how each can be reduced.

[4 marks]

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0 7 . 3 Explain why it is more efficient to have separate collection systems for foul sewage and for surface drainage water.

[2 marks]

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0 8 . 1 Explain how energy absorbed in tropical regions is distributed to higher latitudes such as the UK.

[4 marks]

0 8 . 2 Explain why the UK has milder winters than Moscow or Churchill, Canada, which are at similar latitudes.

[2 marks]

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0 8 . 3 Explain why the atmosphere would not retain liquid water if atmospheric pressure were much lower.

[2 marks]

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0 8 . 4 Give **two** industrially important gases extracted from the atmosphere.

[2 marks]

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0 9 . 1 Describe the main features of the Montreal Protocol (1987).

[3 marks]

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0 9 . 2 For each of the following CFC uses, give the alternative process or material now used and one disadvantage of the alternative.

[4 marks]

Aerosol propellant .....

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Refrigerant .....

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0 9 . 3 Describe how waste CFCs from old refrigerators are disposed of safely.

[2 marks]

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0 9 . 4 State **one** reason why the ozone layer is not expected to recover fully for many decades.

[1 mark]

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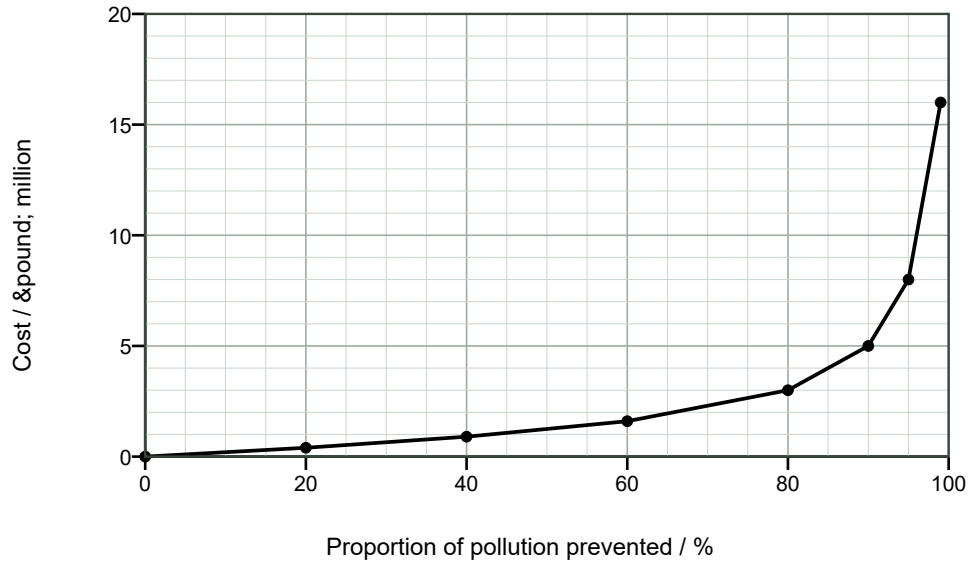
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1 0 Figure 1 shows how the cost of controlling emissions from a factory changes with the proportion of the pollutant that is prevented from being released.

Figure 1



1 0 . 1 Use Figure 1 to calculate the extra cost of increasing control from 90% to 99% of the pollutant.

[1 mark]

Answer ____ £ million

1 0 . 2 Explain how the terms **ALARA** and **BATNEEC** are used when deciding how much pollution control is required.

[3 marks]

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1 0 . 3 Explain why the control methods that were originally adequate for a factory may no longer be sufficient.

[2 marks]

1 0 . 4 Evaluate the principles and general strategies that can be used to decide how pollution should be controlled.

[9 marks]

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Extra space

15

Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

OR

1 1 . 2 To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS